Lecture 16

Dijkstra, Flow Networks

Source: Introduction to Algorithms, CLRS and Kleinberg & Tardos

Dijkstra's Algorithm: Sketch

Maintain a set of explored vertices S for which algorithm has found $d[u] = \delta(s, u)$:

Step 1: Initialise $S = \{s\}$, $d[s] = 0$.

Step 2: Choose an unexplored vertex v from $V \backslash S$ which minimizes:

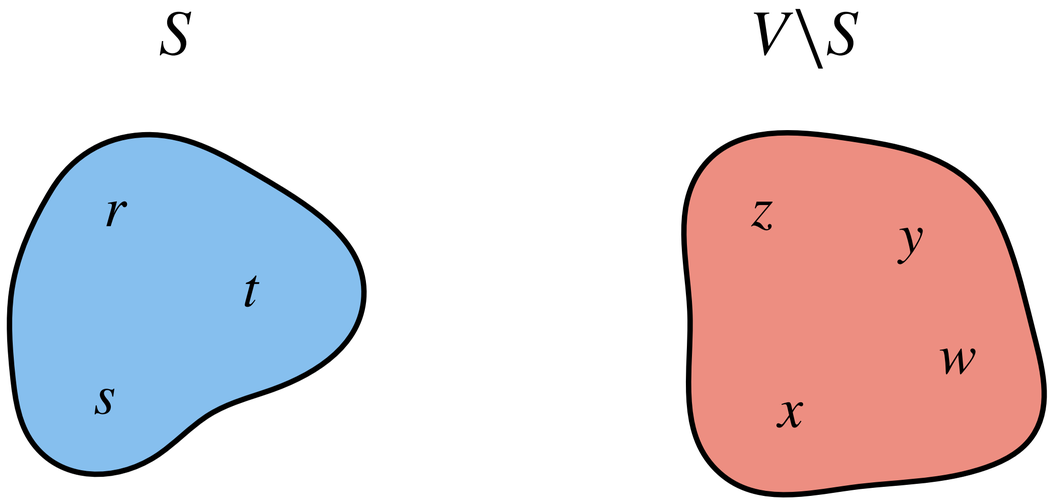
$$\pi[v] = \min_{(u,v) \in E,\ u \in S} d[u] + w(u, v)$$

Add v to S and set $d[v] = \Pi[v]$.

Step 3: Go to **Step 2** if it can be performed.

Dijkstra's Algorithm: Optimization

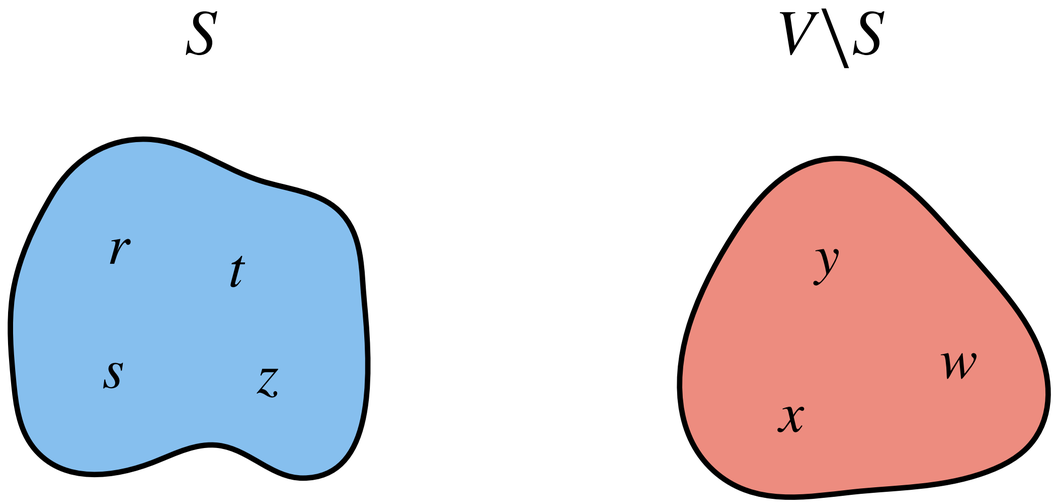
Computed $\pi[v]$ values of $V \backslash S$.



$d[s] = 0, d[r] = 3, d[t] = 2$

$\pi[z] = 4, \pi[y] = 8, \pi[x] = 6, \pi[w] =$ Invalid

Dijkstra's Algorithm: Optimization



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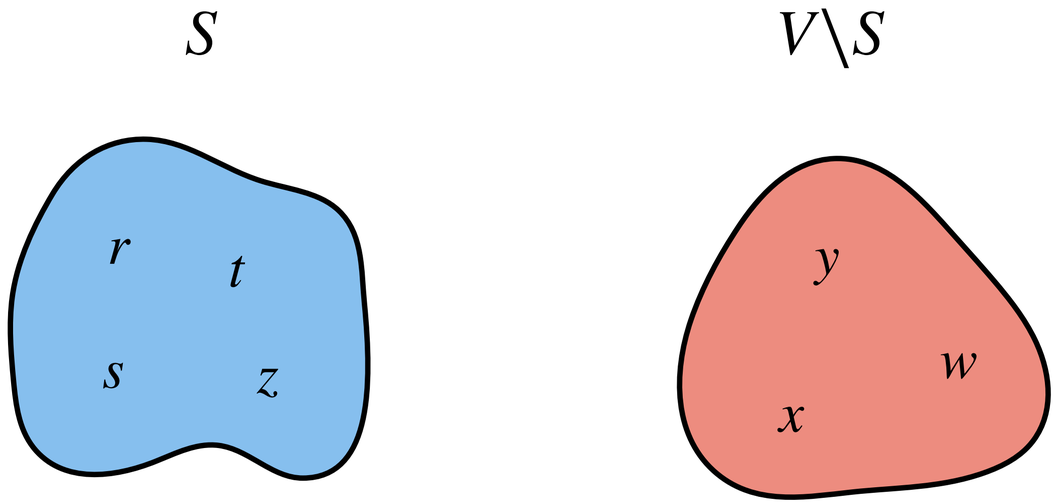
Dijkstra's Algorithm: Optimization

Computing new $\pi[v]$ values after updating S.



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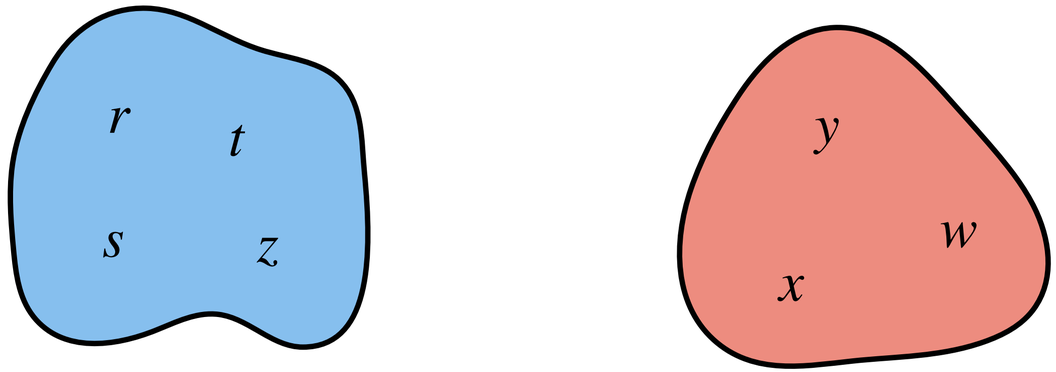

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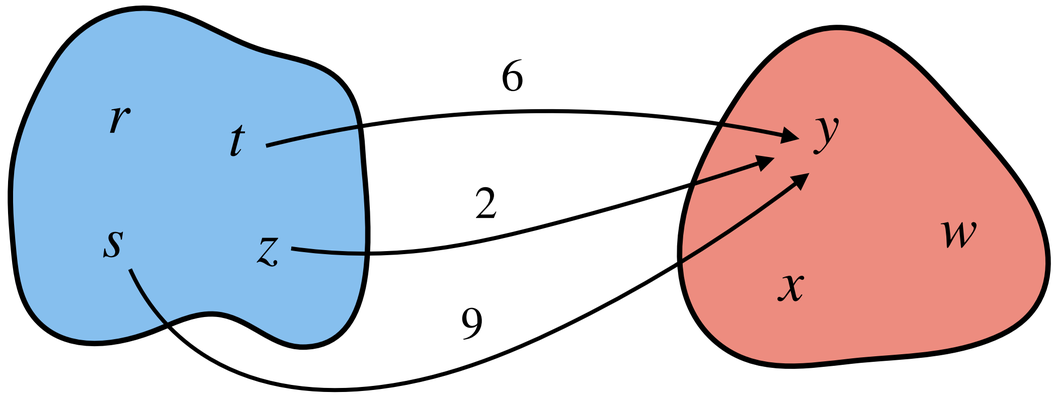

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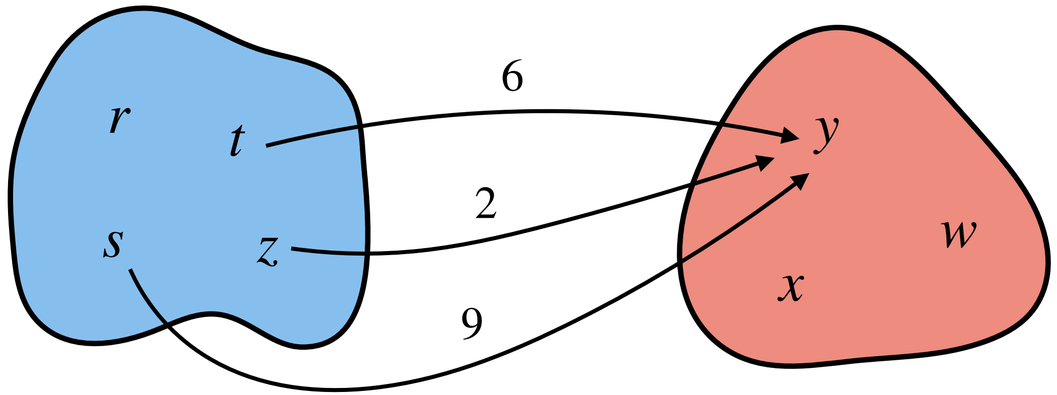

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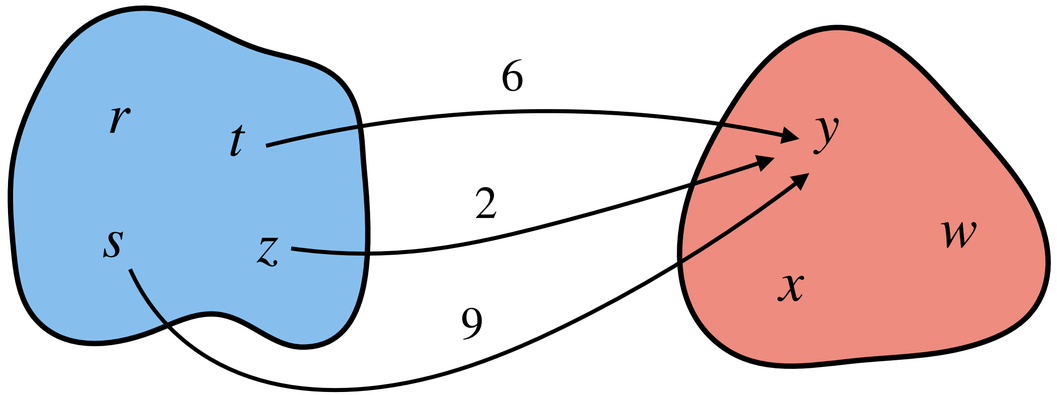

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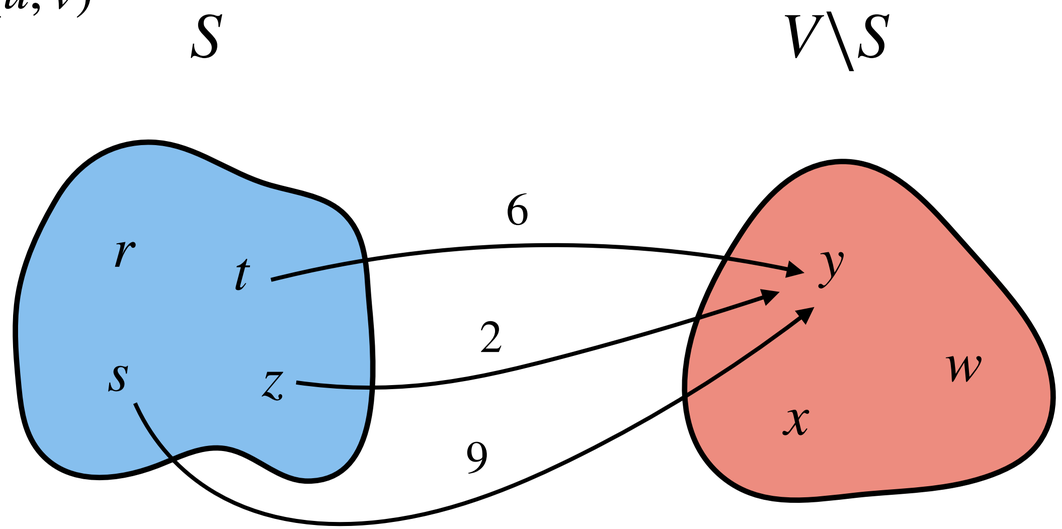

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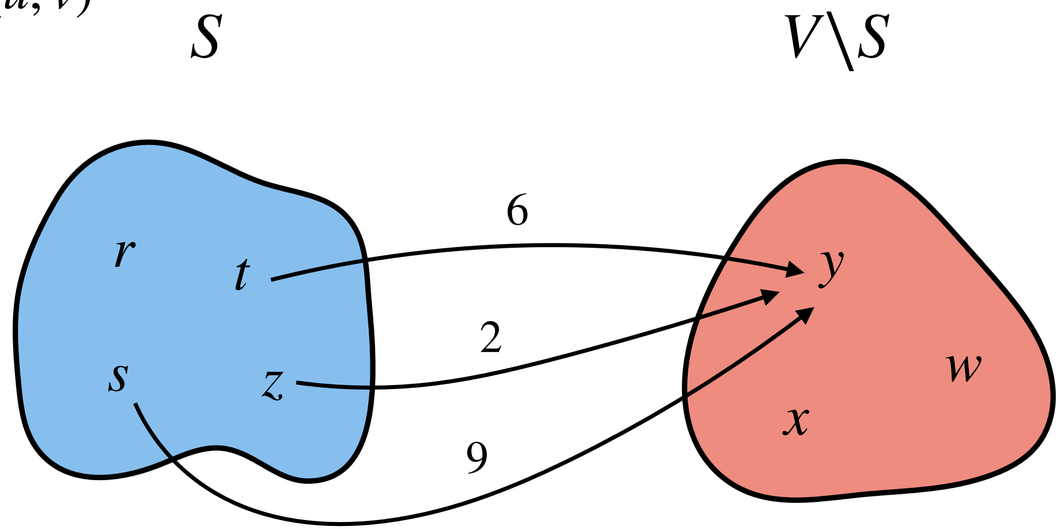

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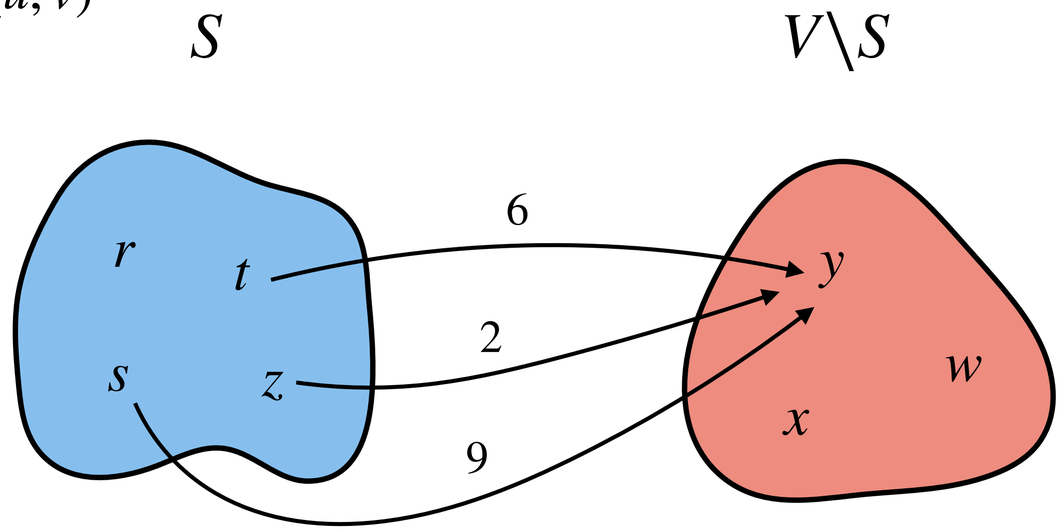

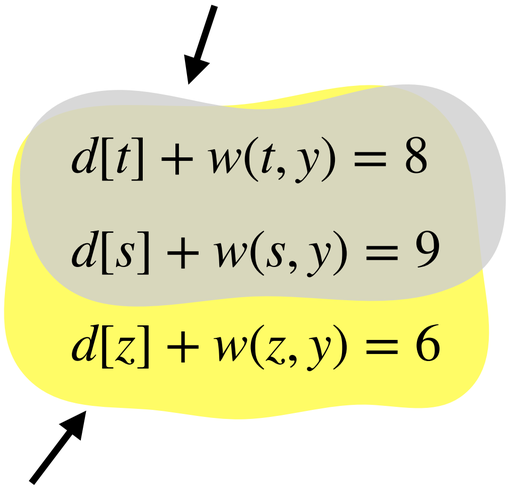

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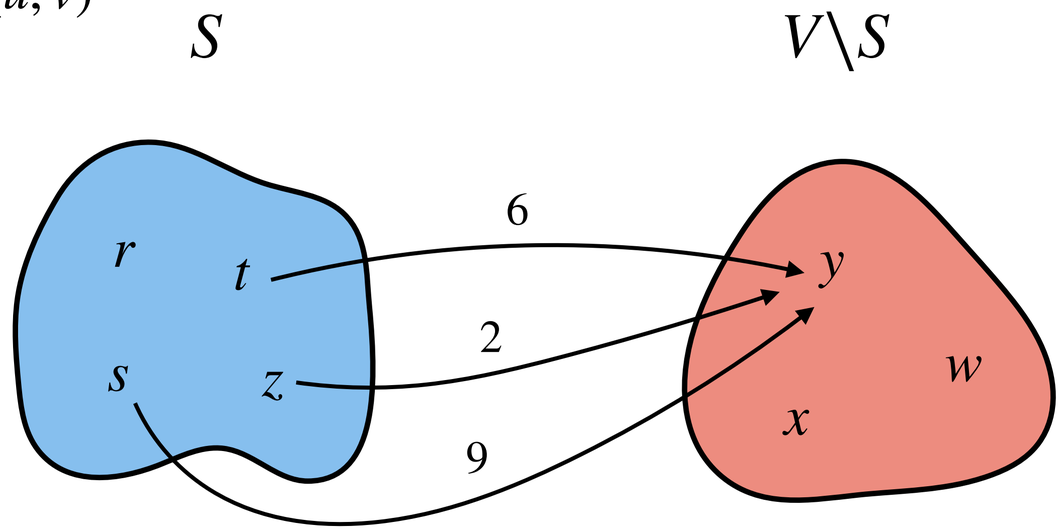

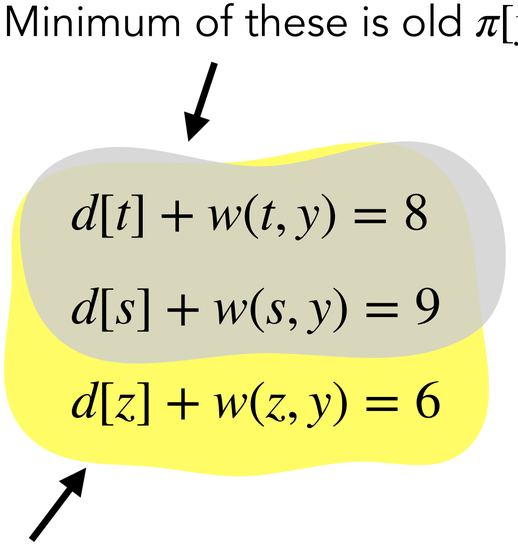

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Conclusion: Form a min-priority queue of $V \setminus S$ where $keys$ are π values.

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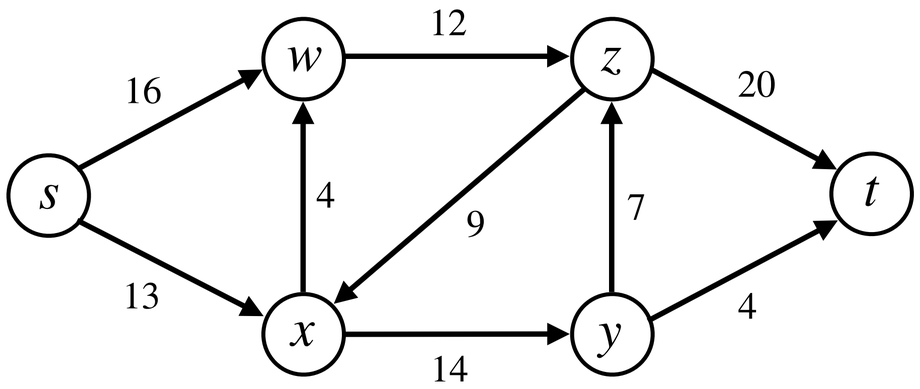
Proof: Already done.

Flow Networks

Flow Networks



(a)

Flow Networks

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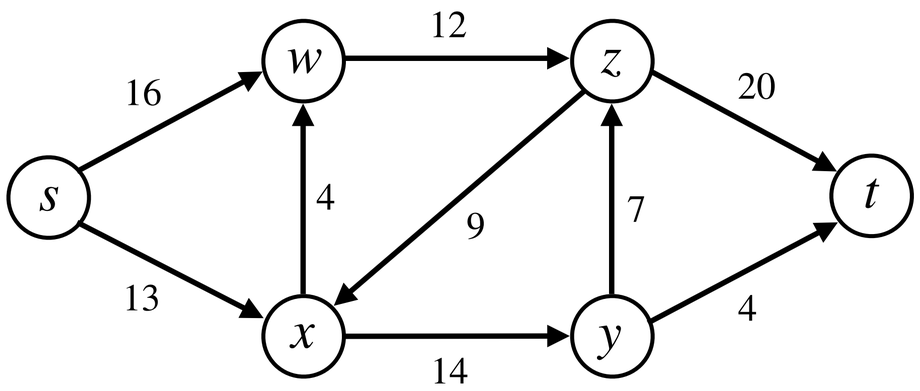

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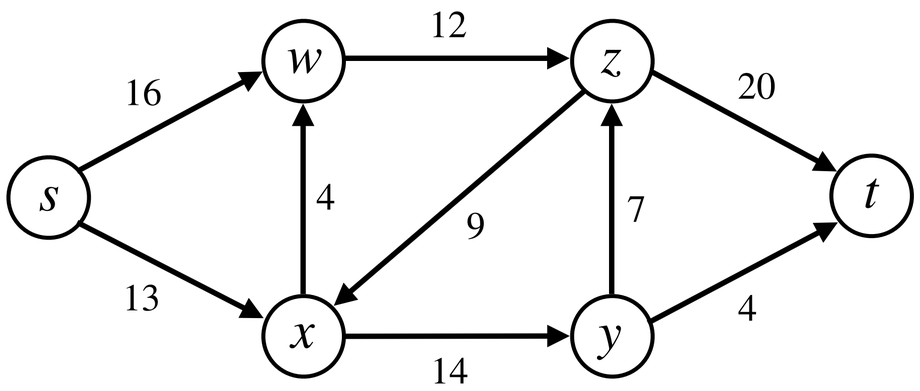

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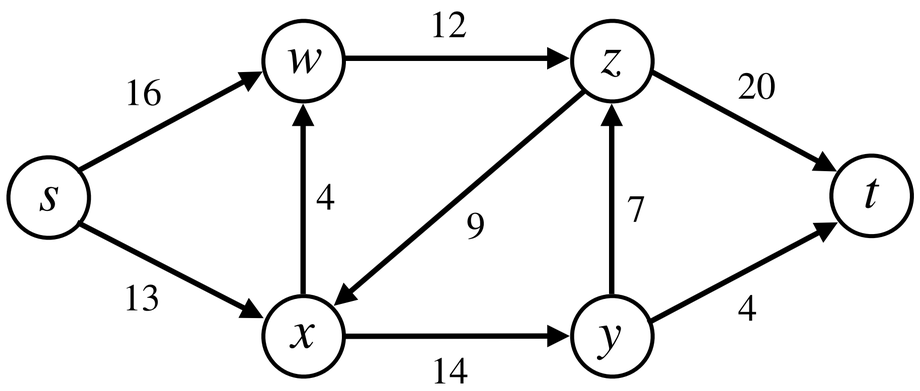

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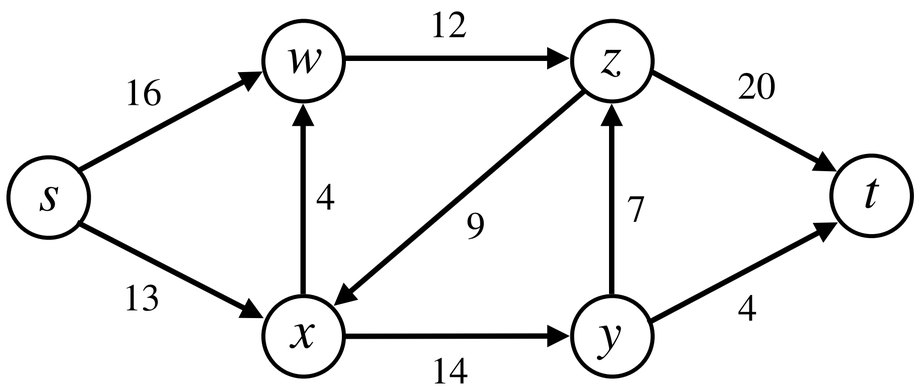

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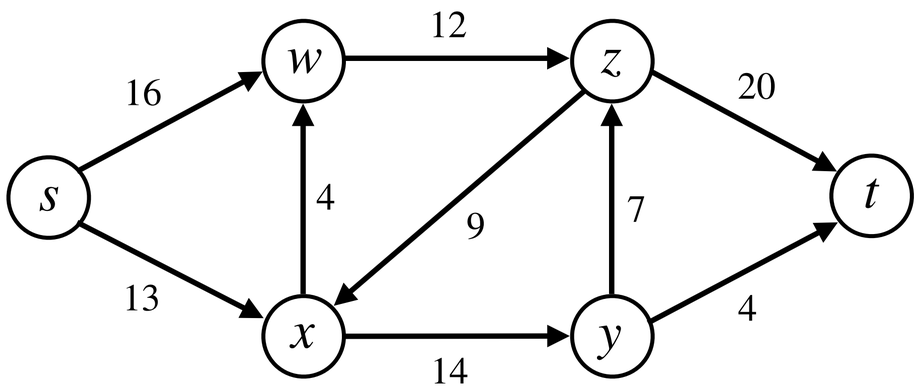

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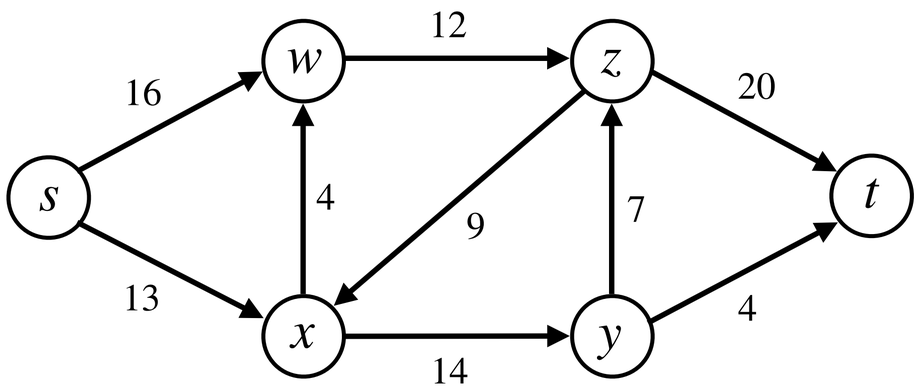

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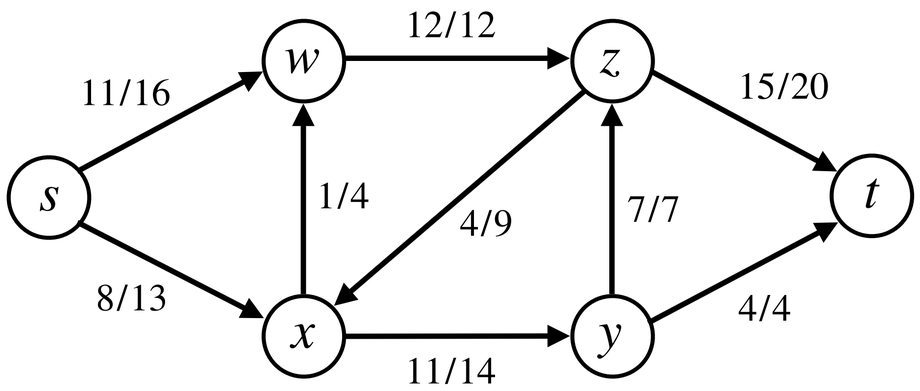

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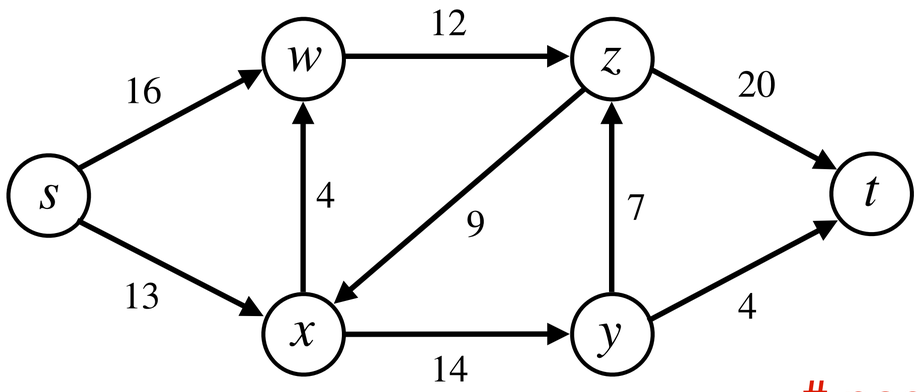

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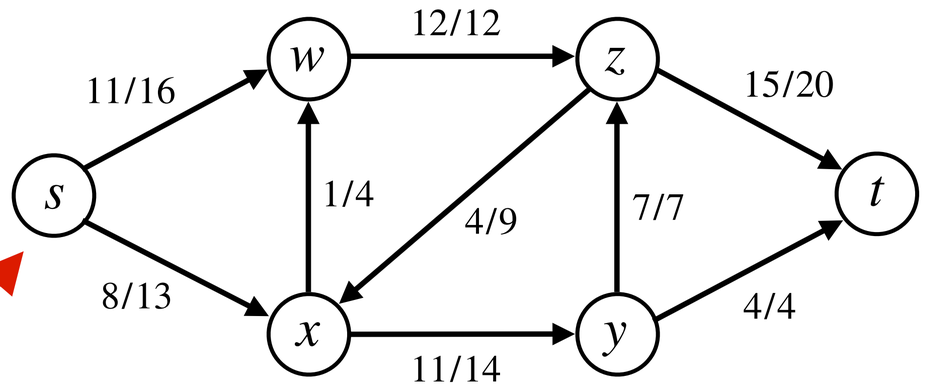

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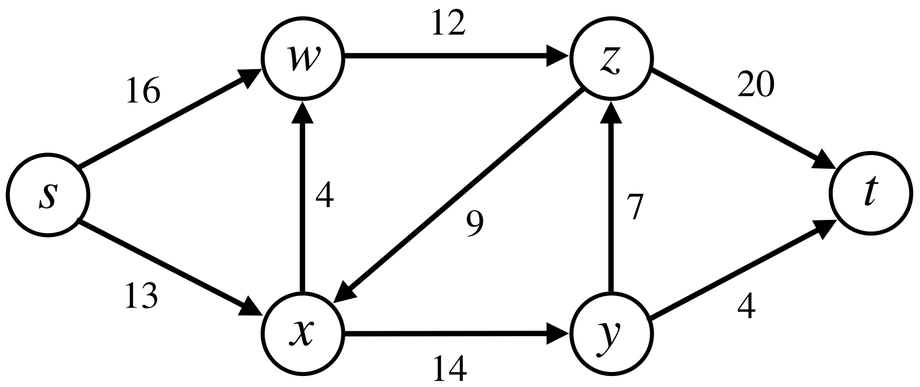

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- **Capacity constraint**: For all $u, v \in V$, $0 \leq f(u, v) \leq c(u, v)$.
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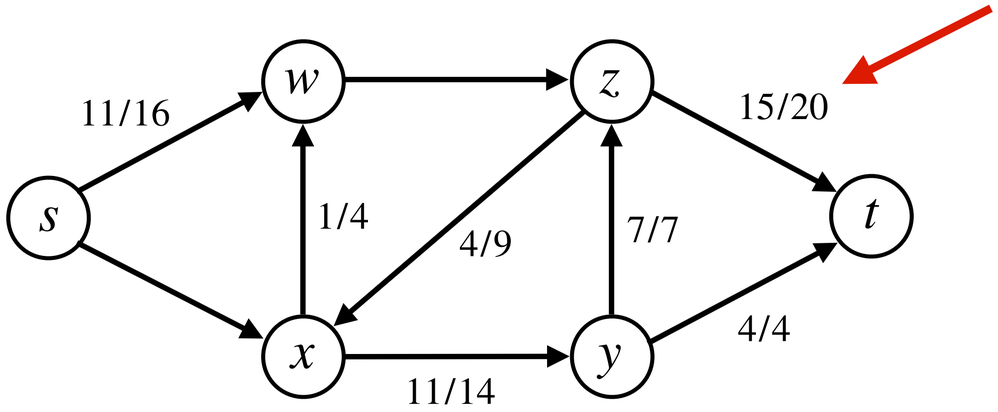

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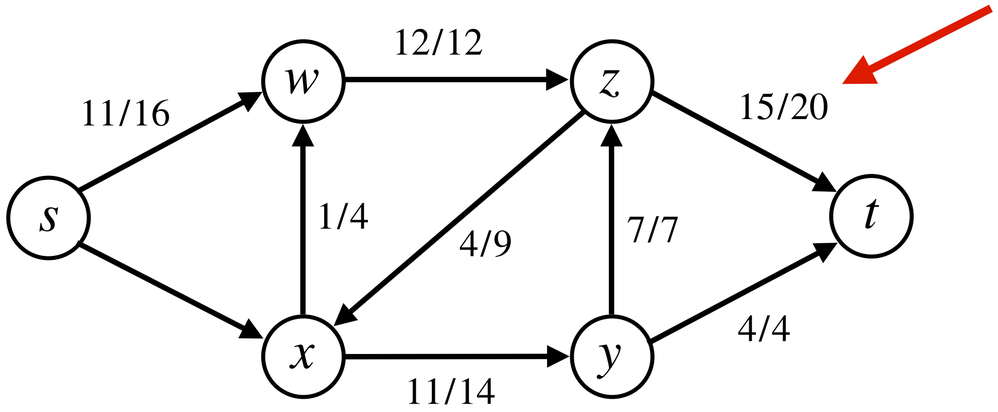

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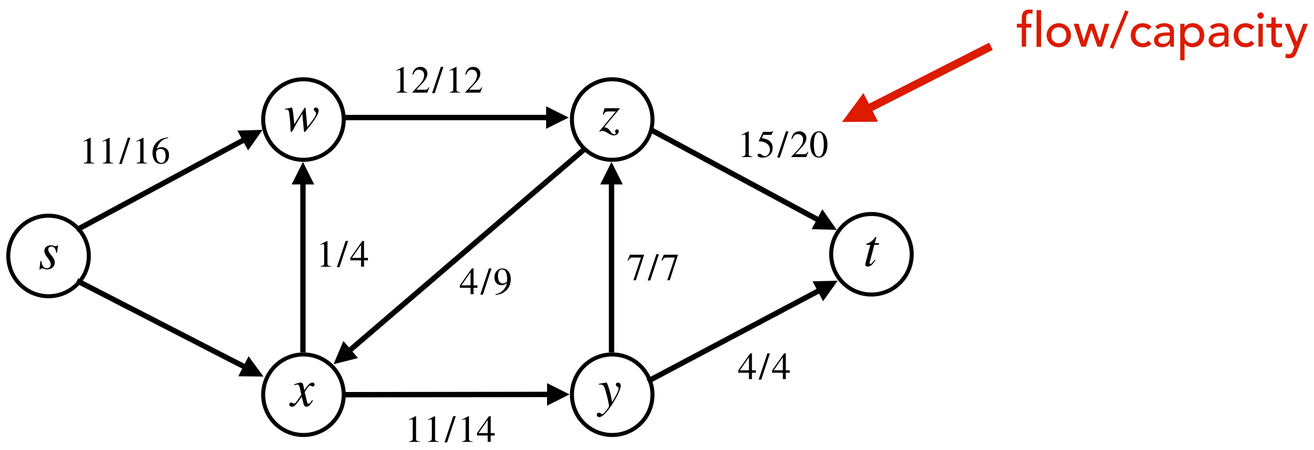

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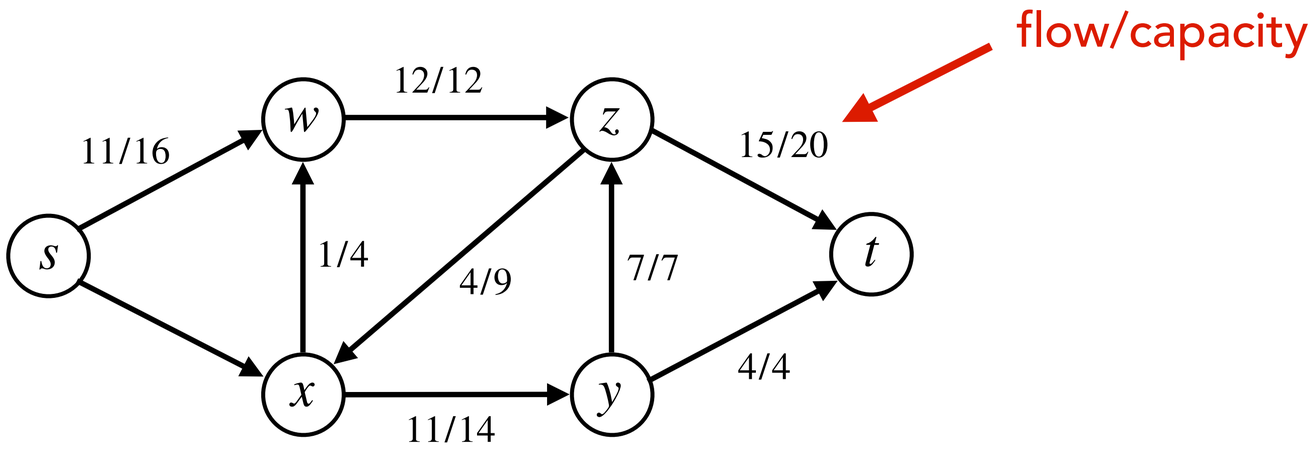
Example:



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Input: A flow network G with source s and sink t.

Output: Find a flow of maximum value.